Prasad Sadanand Mahabare

prasadmahabare8@gmail.com | +918484976195
linkedin.com/prasad-mahabare | github.com/MahabarePS



TECHNICAL SKILLS

- **Languages:** Embedded C, Linux, Python
- **Developer Tools:** VsCode, Keil, MPLabx, STM32CubeIDE, ArduinoIDE
- **Simulation Platforms:** Proteus8, PicSimLab, VMWare
- **Protocols:** UART, I2C, SPI
- **Ticketing Systems:** ServiceNow, Footprints, TeamDynamix
- **Microcontrollers:** 8051,8052,PIC18F4520, STM32F446RE, ESP32-WROOM
- **IoT Cloud Platform:** Blynk, ThinkSpeak, RedNode

PROFESSIONAL EXPERIENCE

Student Network Engineer, Purdue University **Jan 2023 – Dec 2023**

- Configured and installed Cisco catalysts (2650s to 9300s), firewalls, load balancers, DNS.
- Maintained Purdue Cloud System, performance by applying hot fixes and security configurations.
- Developed scripts to automate firmware upgrades over 800 network devices.

EDUCATION

P.G Diploma in Embedded Systems Design **Sep 2025 - May 2026**
BICARD Institute, Pune

Master of Science in Computer Science **Jan 2022 - Dec 2023**
Purdue University, Fort Wayne, Indiana

Bachelor of Engineering Computer Engineering **Aug 2016 - Aug 2019**
Pune University, Pune, India

PROJECTS

Simple Emulator Study

- Constructing an **8-bit CPU emulator** in C focused on **opcode interpretation**, **register management**, and **memory mapping**.
- **Simulating instruction execution cycles** and **stack behavior** to strengthen computer architecture fundamentals.

ESP32 IoT & Firmware Development

- Implemented ESP32-based applications involving **analog input acquisition**, **PWM output control**, and peripheral interfacing.
- Explored wireless-enabled embedded systems using GPIO handling, **sensor integration**, and **real-time processing** concepts.

STM32 Embedded Systems

- Configured STM32 **ARM Cortex-M4** peripherals including **ADC, PWM, UART, SPI, I2C,** and **interrupt** modules.
- Analyzed reference manuals and implemented **firmware** using **HAL, LL,** and **register-level programming techniques**.

PIC Embedded System Development

- Created firmware applications on PIC microcontrollers for **timer operations**, **serial communication**, and **signal generation**.
- Simulated and validated hardware interfaces such as LEDs, push buttons, and **LCD** modules using **MPLAB X** and Proteus.